Right Tech, Wrong Time.

How to determine if your ecosystem is ready for the next technological revolution.

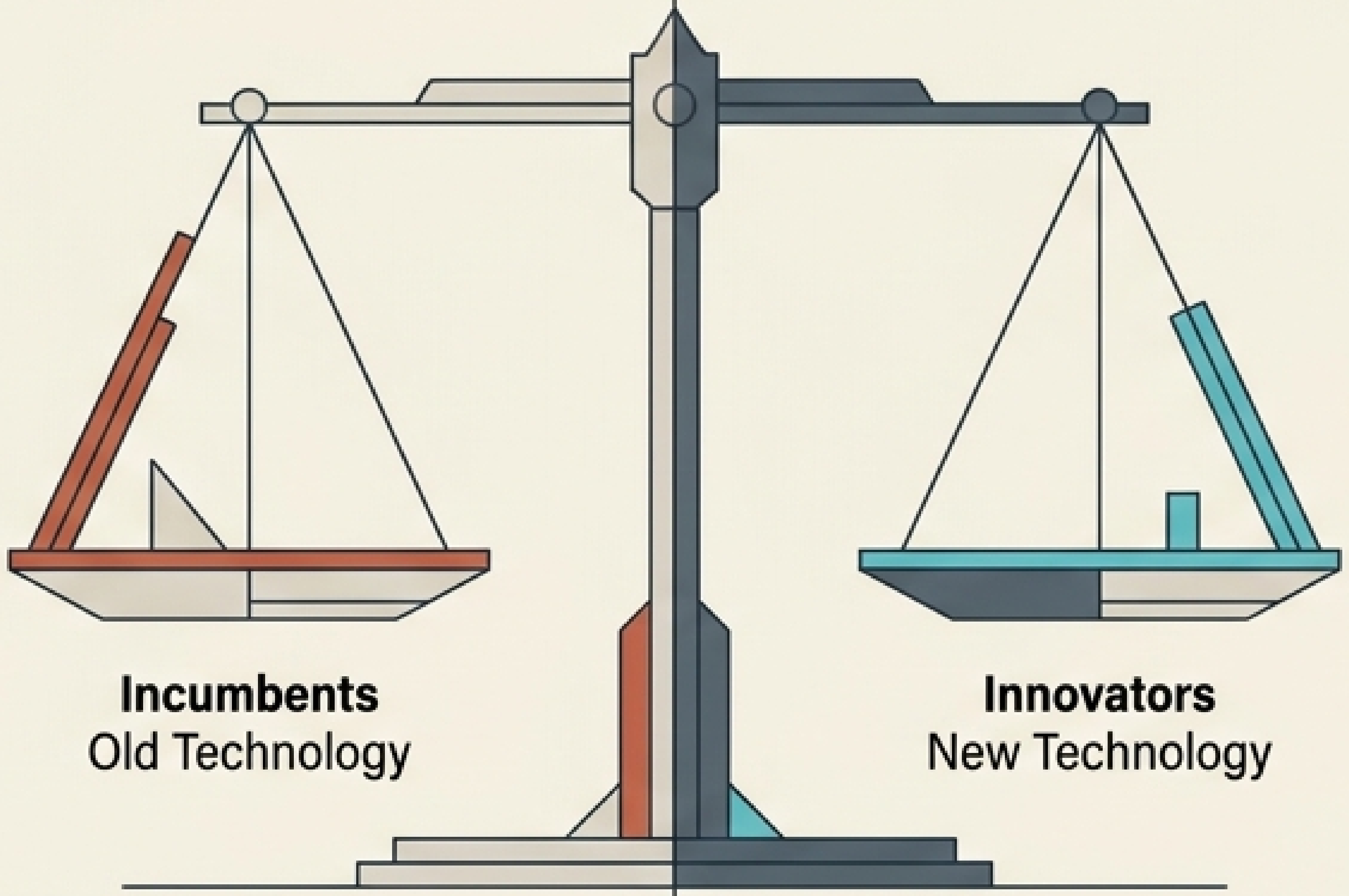
Based on the insights of Ron Adner and Rahul Kapoor.

We excel at predicting the threat of innovation, but fail at predicting its timing.

The First Fear: Being Ready Too Late

Missing the revolution entirely while clinging to an outdated model.

Example: Blockbuster ignoring the shift from video rentals to streaming.



The Second Fear: Getting Ready Too Soon

Exhausting resources and capital before the market is actually ready to transition.

Example: The early dot-com pioneers who died in the 2001 crash, only to see their exact ideas reborn as profitable Web 2.0 ventures years later.

The timing of technological change remains a mystery.
To solve it, we must stop looking exclusively at the technology itself.

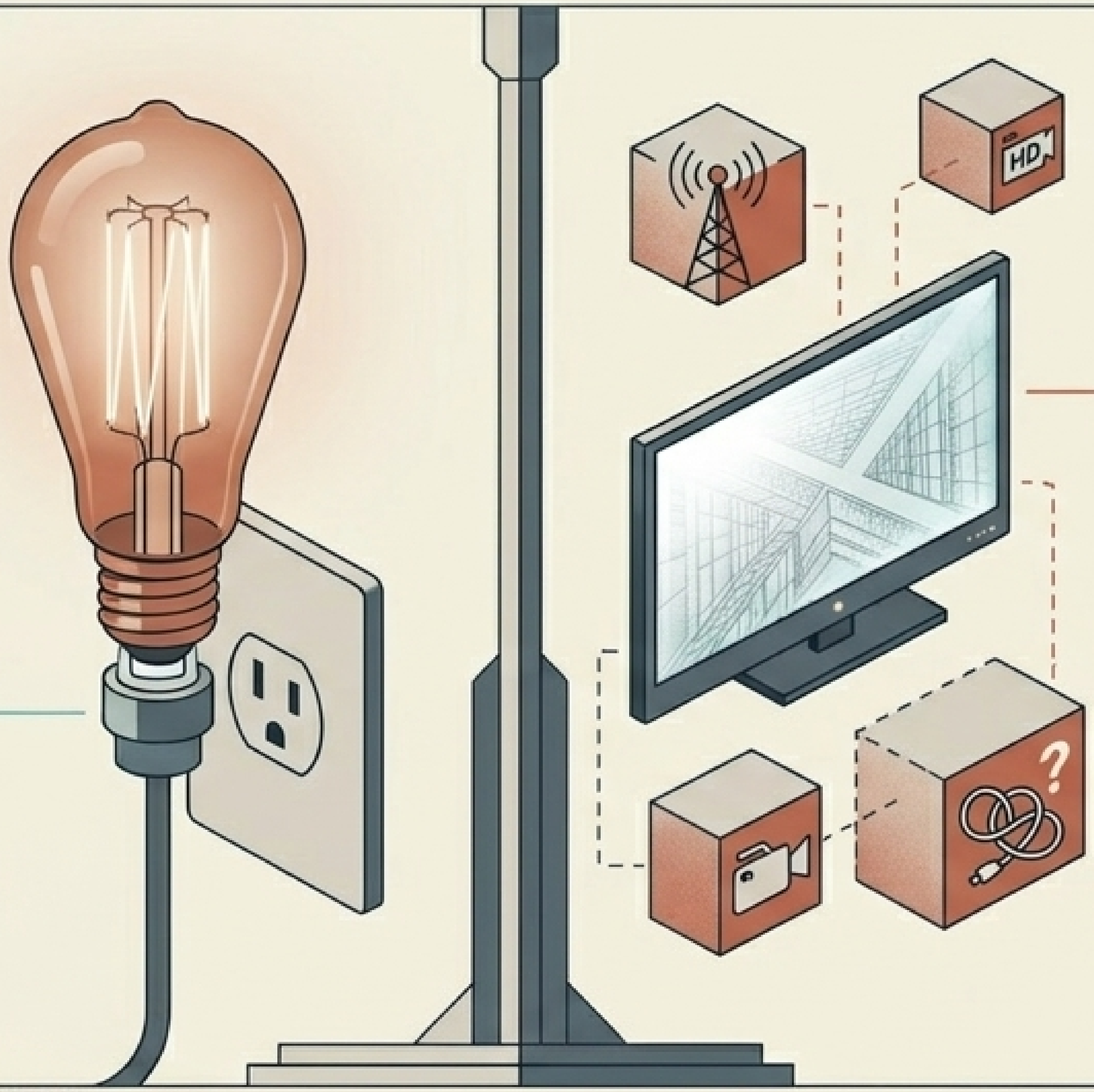
A breakthrough technology is only as strong as its surrounding ecosystem.

Plug-and-Play

The Lightbulb Model.

The value proposition does not hinge on external factors. It plugs into the existing grid.

Great product execution translates immediately into great results. Adoption can be swift.



High Interdependence

The HDTV Model.

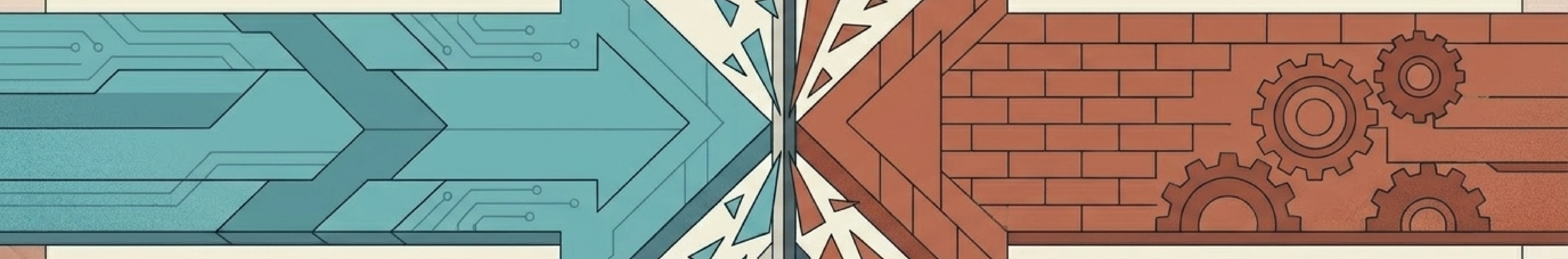
Value creation is blocked until co-innovations are commercially viable.

HDTV pioneers had the right vision in the 1980s, but waited 30 years for high-def cameras, new broadcast standards, and updated post-production processes to emerge.

Substitution is a race between warring ecosystems.

New Tech's Emergence
Challenge

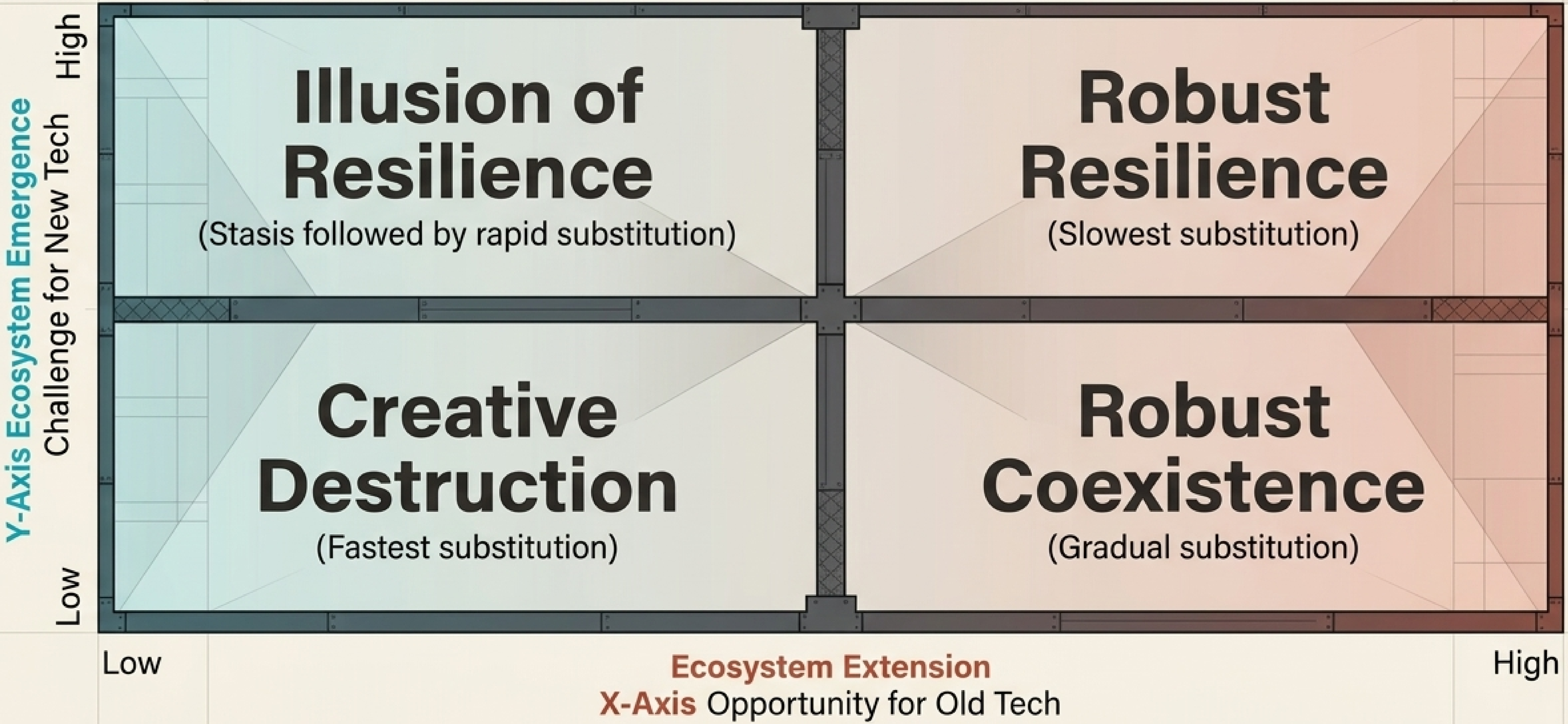
Old Tech's Extension
Opportunity



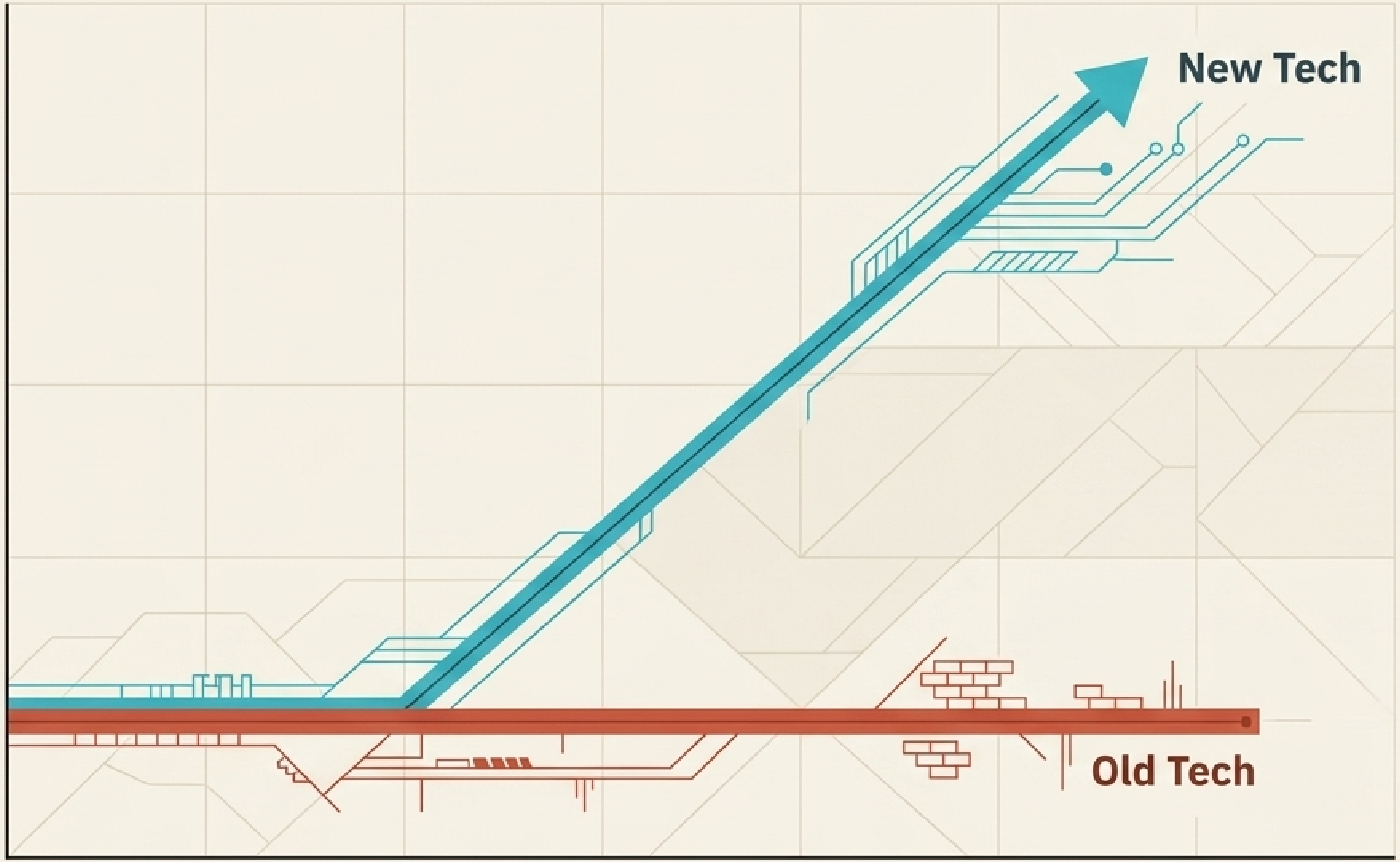
How quickly can the new ecosystem resolve bottlenecks? (e.g., Cloud computing relying on the deployment of widespread broadband and online security).

How much can the incumbent technology improve its competitiveness without changing its core? (e.g., DSL technology pushing 15 megabytes per second through ancient copper telephone lines to fight off fiber networks).

The four quadrants of technological change.



Creative Destruction delivers the fastest market substitution.



The Conditions

New Tech Challenge is LOW.
Old Tech Opportunity is LOW.

The Dynamics

The new technology's value is not held back by bottlenecks, and the old technology has zero potential to improve in response.
The upstart swiftly causes the demise of established competitors, relegating the old tech to tiny niches.

Historical Precedent

16GB flash drives replacing 8GB flash drives.

Inkjet printers rapidly obliterating Dot Matrix printers.

Robust Coexistence results in a prolonged, dual-market battle.

The Conditions

New Tech Challenge is LOW.
Old Tech Opportunity is HIGH.

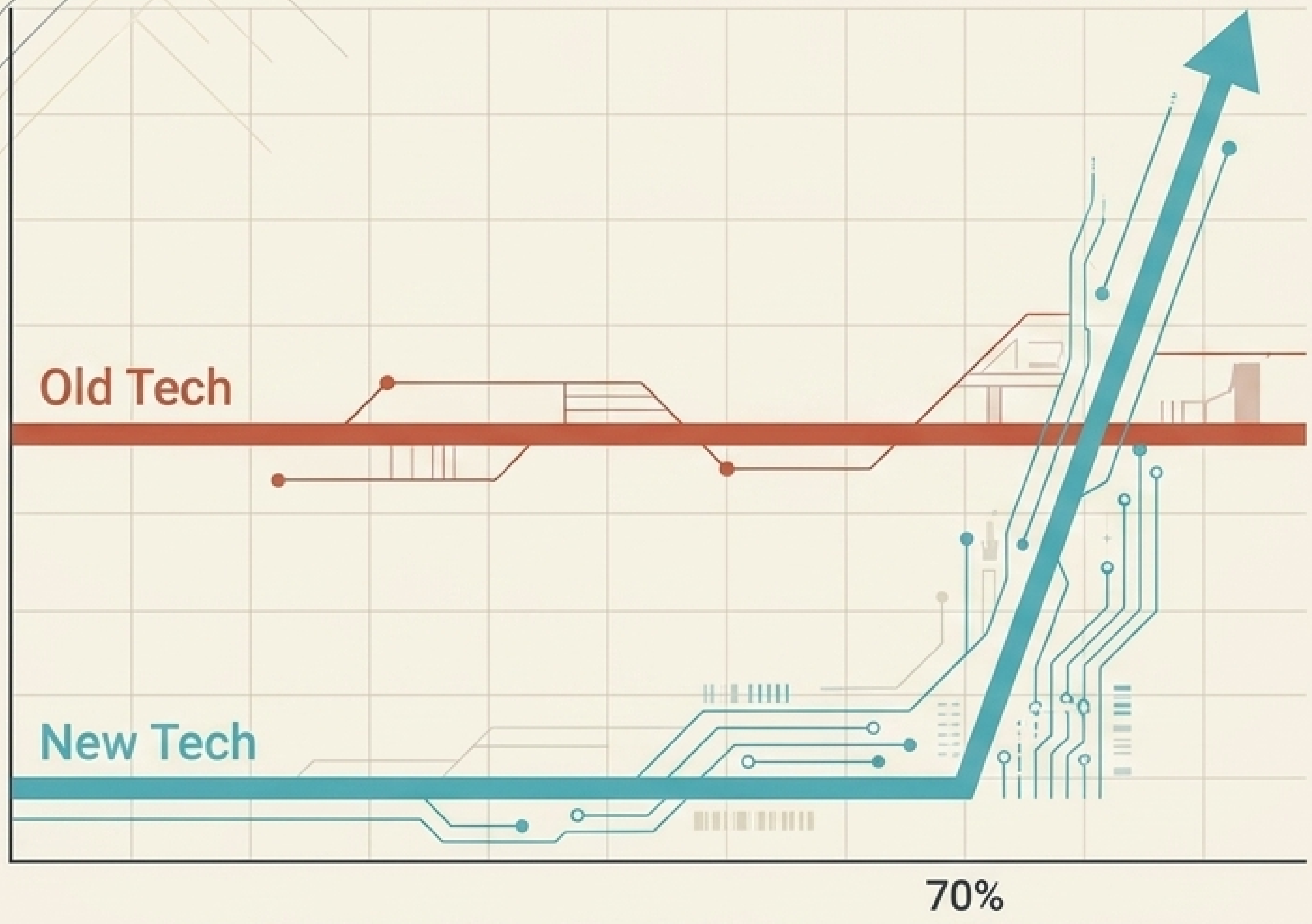
The Dynamics

The new technology makes immediate inroads because it is compatible with the existing ecosystem.
However, the old technology successfully defends market share by aggressively improving its own ecosystem.

Historical Precedent

Hybrid engines vs. Internal-Combustion engines. (Hybrids don't require new charging grids, but traditional gas engines continue to become drastically more fuel-efficient).

The Illusion of Resilience masks fragile incumbent dominance.



The Conditions

New Tech Challenge is HIGH.
Old Tech Opportunity is LOW.

The Dynamics

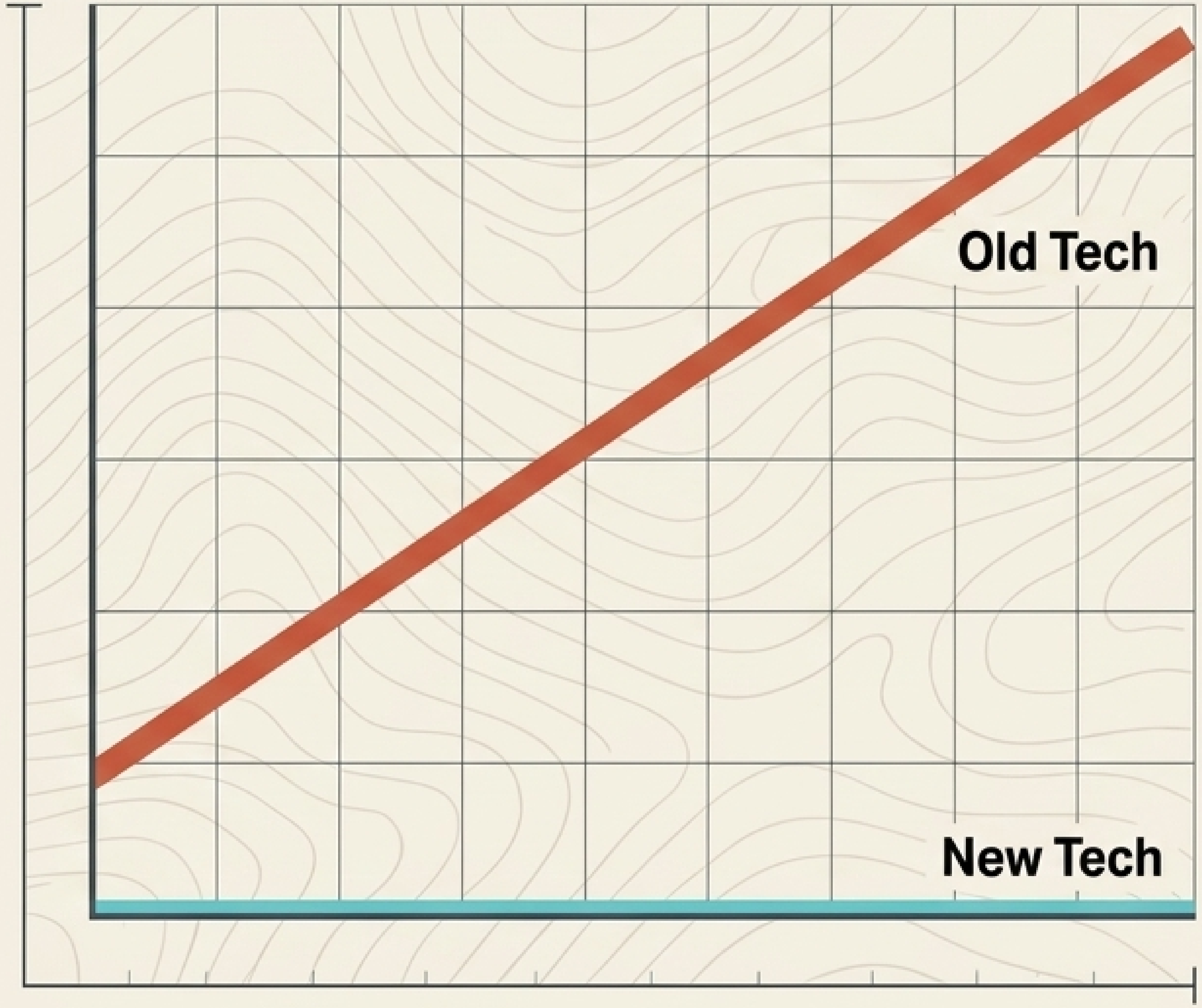
Incumbents maintain high market share, but growth is completely stalled. Their dominance is an illusion—maintained only by unresolved bottlenecks holding back the new tech.

Once the new ecosystem is ready, the market-share inversion is violent and rapid.

Historical Precedent

E-books vs. Printed books.
GPS navigators vs. Paper maps.

Robust Resilience proves that being ten years too early is worse than missing the revolution.



The Conditions

New Tech Challenge is HIGH.
Old Tech Opportunity is HIGH.

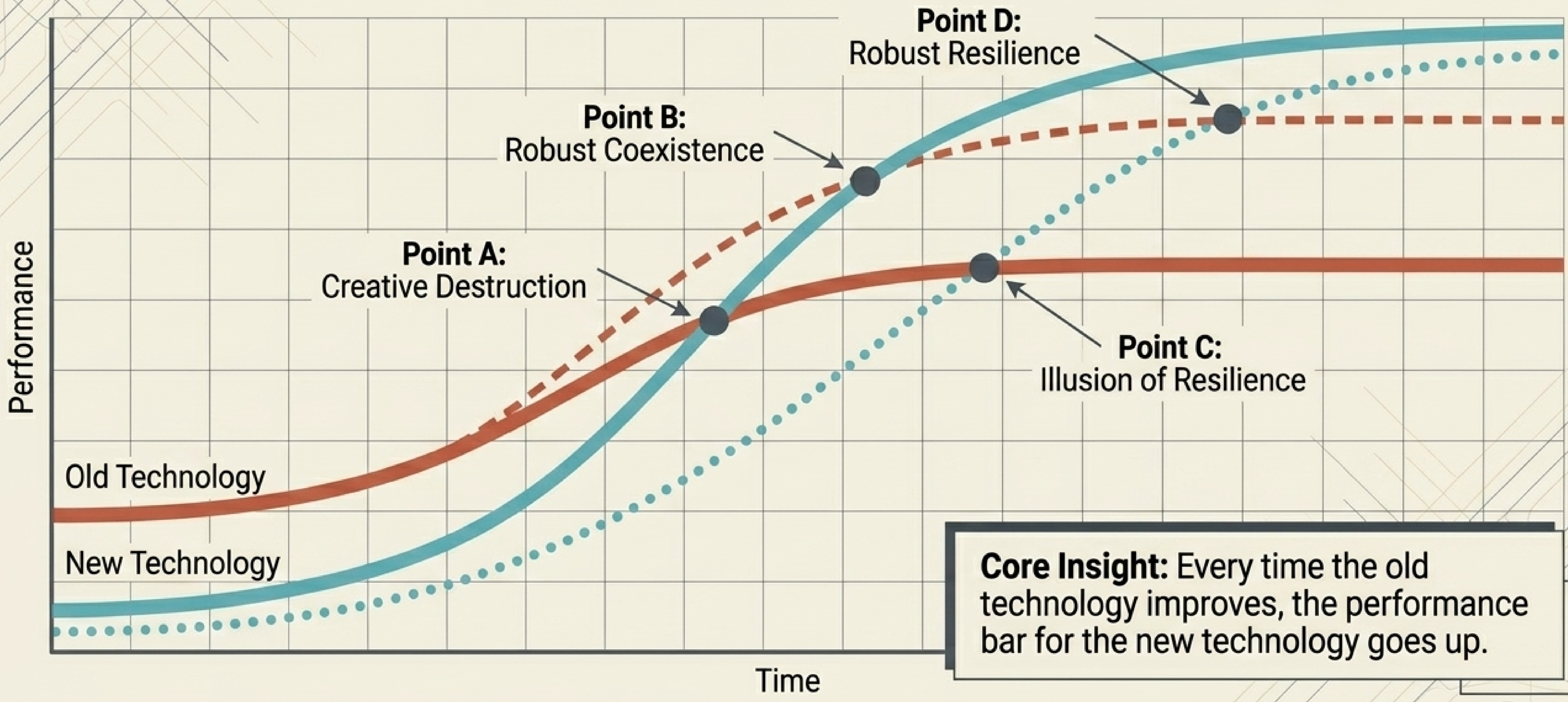
The Dynamics

The old technology maintains a prosperous leadership position for an extended period by constantly upgrading. The new tech appears overhyped and is relegated to narrow niches, waiting for its ecosystem to mature.

Historical Precedent

Bar codes vs. RFID chips. RFID promises far richer data, but its adoption lagged due to slow IT infrastructure deployment. Meanwhile, basic bar codes became vastly more powerful as underlying IT systems enabled real-time tracking.

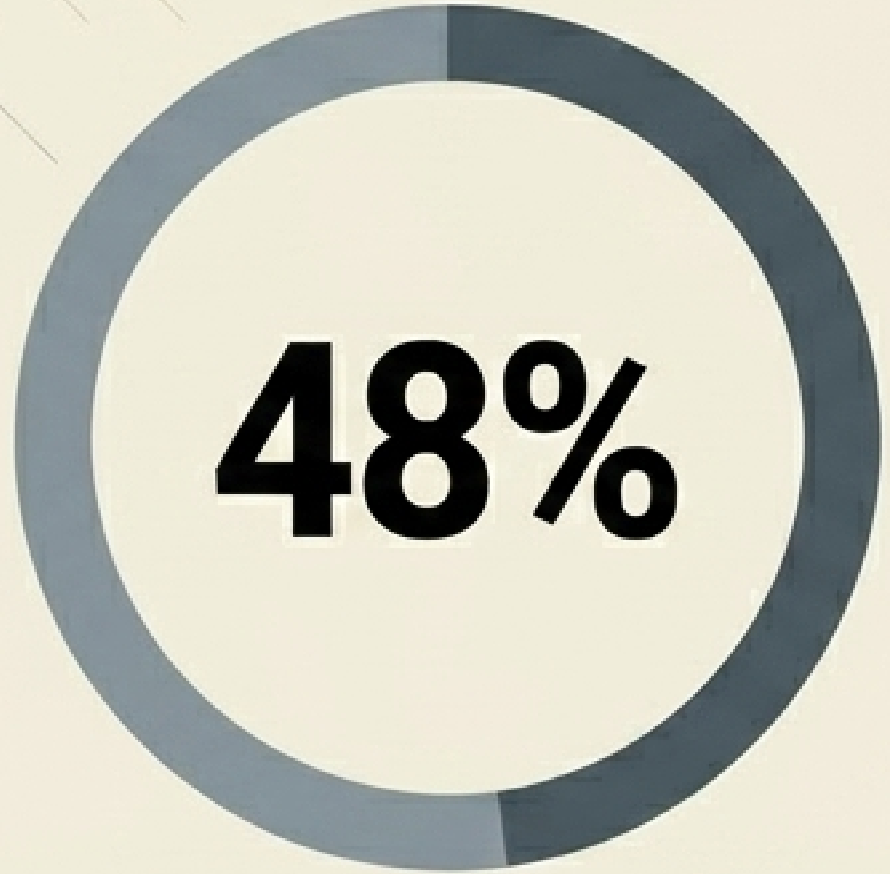
Visualizing the trajectories of substitution.



Core Insight: Every time the old technology improves, the performance bar for the new technology goes up.

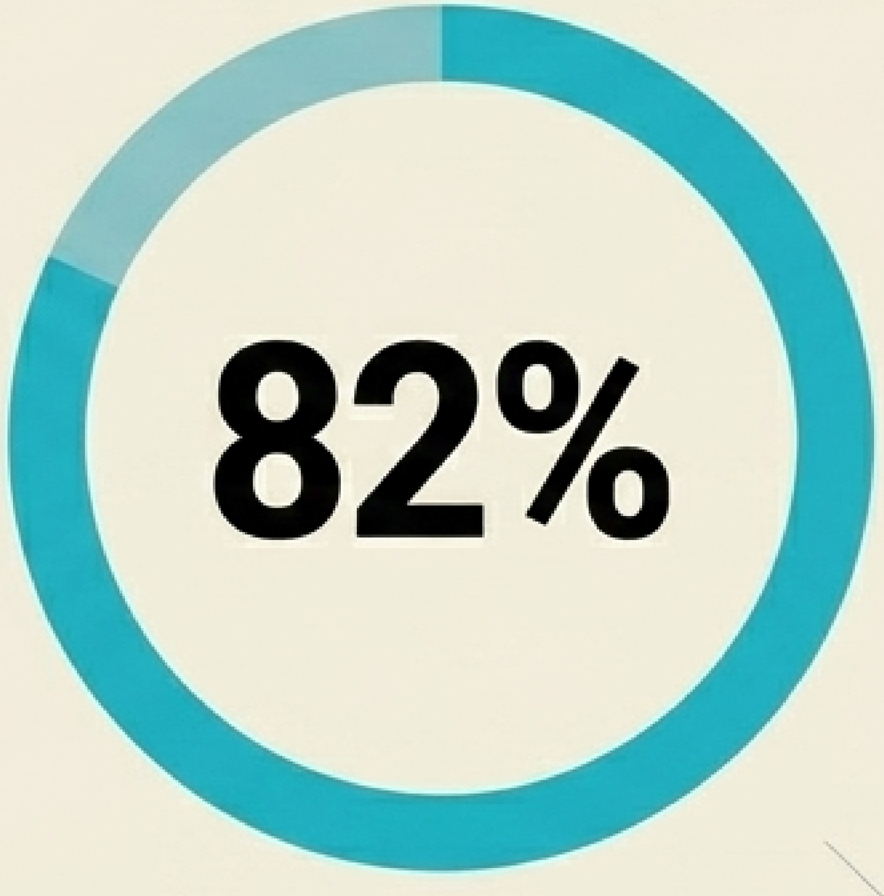
The statistical proof of ecosystem dynamics.

Traditional View



Tracking traditional factors alone (price-adjusted performance, number of rivals, tenure of old tech) only explained 48% of the variance in substitution speed.

Ecosystem View



When ecosystem emergence challenges and extension opportunities were added to the model, the ability to predict the pace of substitution **nearly doubled** to **82%**.

Based on a five-year study analyzing 60 years of semiconductor lithography substitutions to discover what **actually** predicted the pace of market takeover.

The Ecosystem Diagnostic Checklist

Assessing New Tech Risks (Emergence)	Assessing Old Tech Prospects (Extension)
Execution Risk: How difficult is it to deliver the focal innovation to market on time and to spec?	**Core Evolution:** Can the incumbent technology be pushed further through material or design improvements?
Co-innovation Risk: To what extent does success depend on the commercialization of other breakthrough innovations?	**Complementary Boost:** Can the old tech be extended by upgrading the complementary elements in its current ecosystem?
Adoption-chain Risk: Do external partners need to adopt and adapt to the technology before the end consumer can even see the value?	**Strategic Borrowing:** Can the incumbent borrow or integrate innovations from the new technology's ecosystem to stay relevant?

The Resource Allocation Playbook

	Innovator Strategy	Incumbent Strategy
Creative Destruction	**Invest aggressively**. Push for rapid market capture.	**Brace for impact.** Abandon the core; seek long-term survival in highly specialized niche applications.
Robust Coexistence	**Perfect the ecosystem**. Test and refine offerings with early, highly receptive adopters.	**Invest in old tech**. Aggressively fund improvements to your existing ecosystem to delay the transition.
Illusion of Resilience	**Divert resources away** from perfecting the core technology and strictly toward fixing external ecosystem bottlenecks.	**Harvest current profits and prepare for sunset.** Do not over-invest in the old technology; the end will be sudden.
Robust Resilience	**Exercise extreme patience.** Accept that the performance threshold is rising daily. Focus on viable niches until the ecosystem matures.	**Invest aggressively.** Actively raise the performance bar that challengers are forced to cross.

The Right Tech Requires the Right Time

1

Look Beyond the Product

When the bottleneck to adoption is the ecosystem, pushing technological progress is pulling the wrong lever.

2

Respect the Incumbent

Do not underestimate the old technology's ability to evolve. Every time the old ecosystem improves, the performance bar for your innovation rises.

3

Sequence Your Strategy

Predicting if an innovation will win is strategy. Predicting when the ecosystem will allow it to win is survival. Allocate your resources accordingly.